Rankings: Slope, Bump & Waterfall

Workshop 3 — Module 6

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

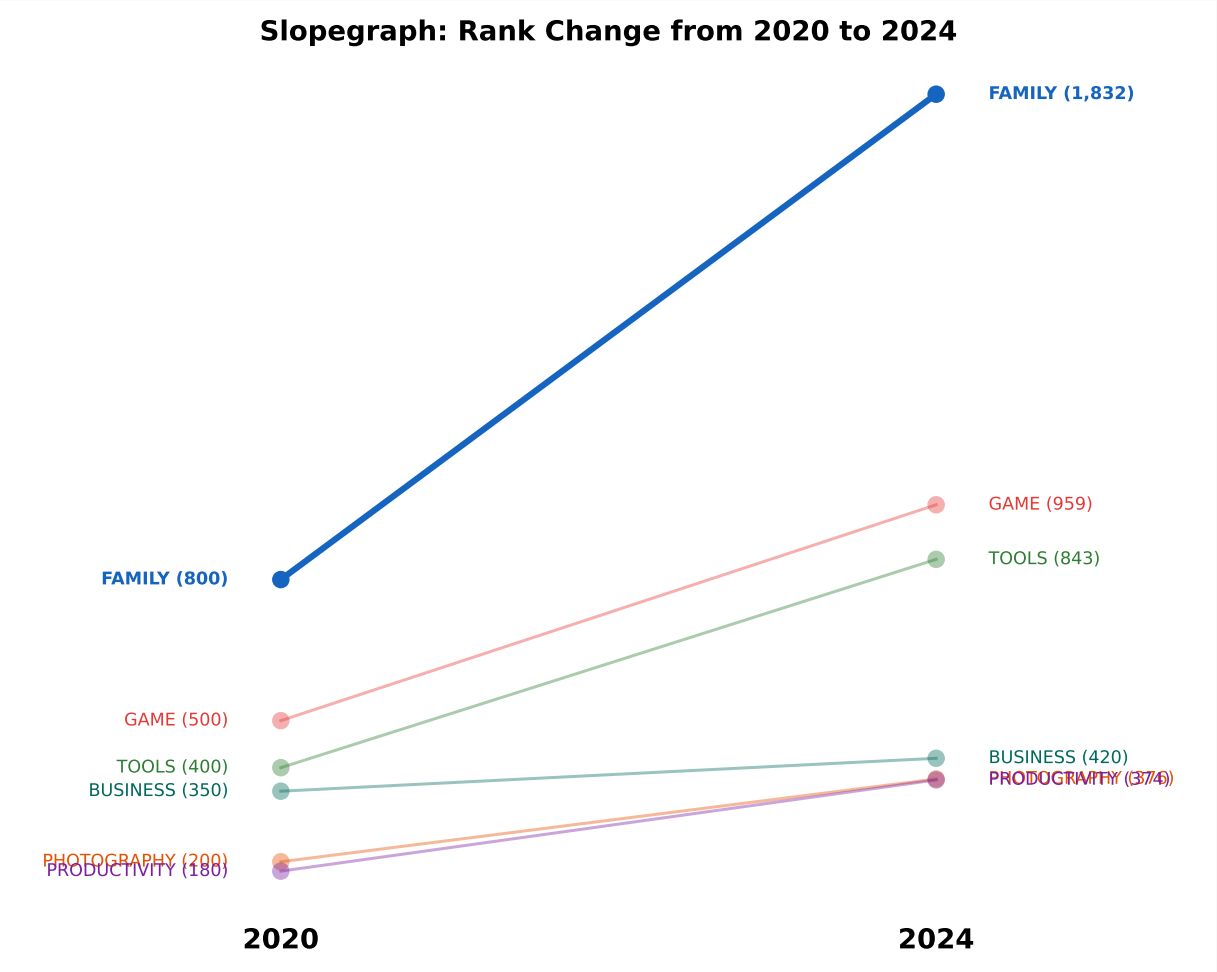
1. Slopegraphs: two-point before/after comparison
2. Bump charts: rank trajectories over multiple periods
3. Waterfall charts: sequential cumulative breakdown
4. Grey-accent strategy for slope and bump readability
5. Chart selection: slope vs bump vs waterfall

Learning Outcome

You will produce slopegraphs for before/after comparisons, bump charts for rank trajectories, and waterfall charts for financial P&L breakdowns — three specialised chart types not covered by standard bar or line charts.

Slopegraphs

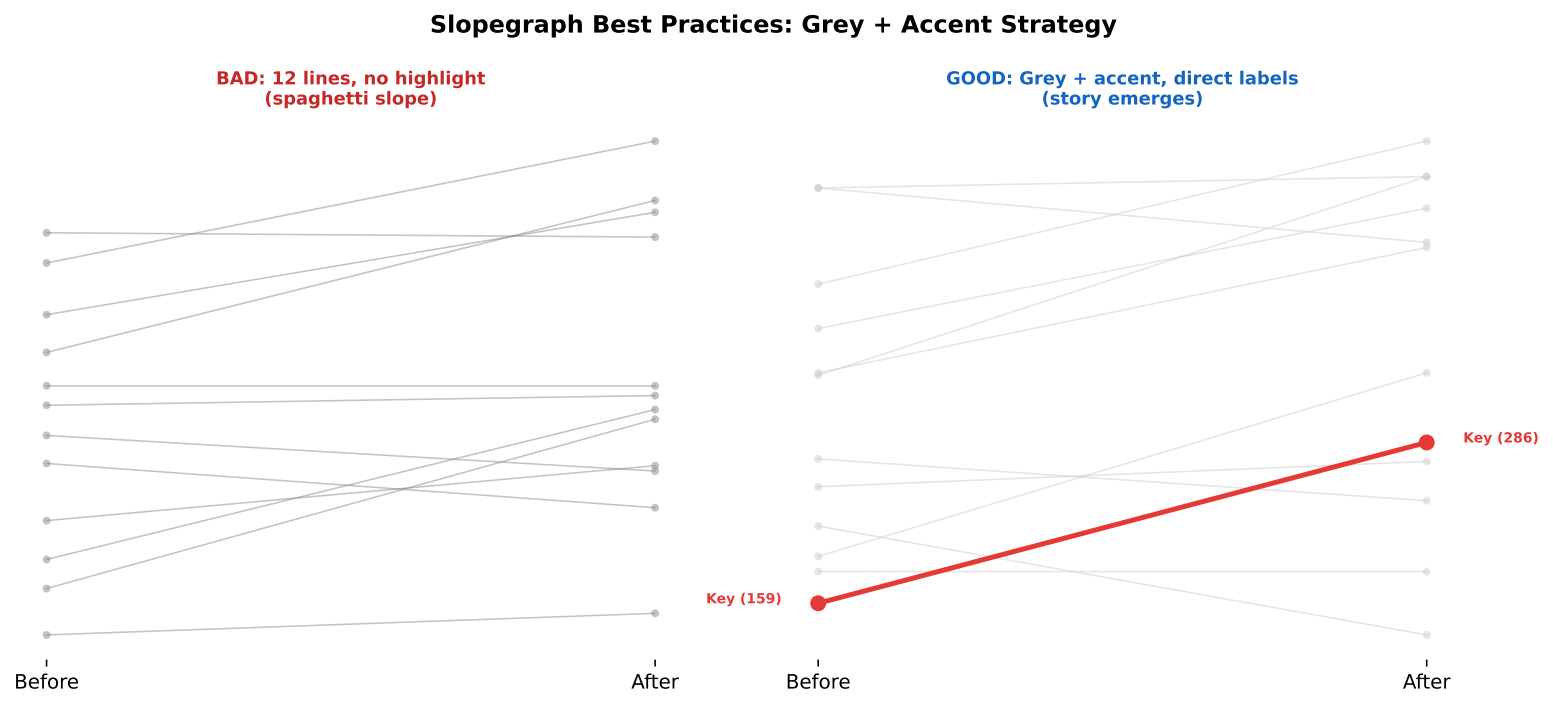
Slopegraph: Two-Point Comparison



Data Insight

A slopegraph (Tufte, 1983) connects two values per item across two time points. Slope encodes rate of change; position encodes absolute value; crossing lines reveal rank changes. Use for 5–15 items across exactly two

Slopegraph Best Practices



Pro-Tip

With >8 lines, slopegraphs become spaghetti. Apply the **grey-accent strategy**: draw all lines in light grey, then highlight 1–2 key items in colour with direct labels at both ends. This is the same technique used for scatter (W01-M04) and line charts.

Slopegraph in R vs Python

R (ggplot2)

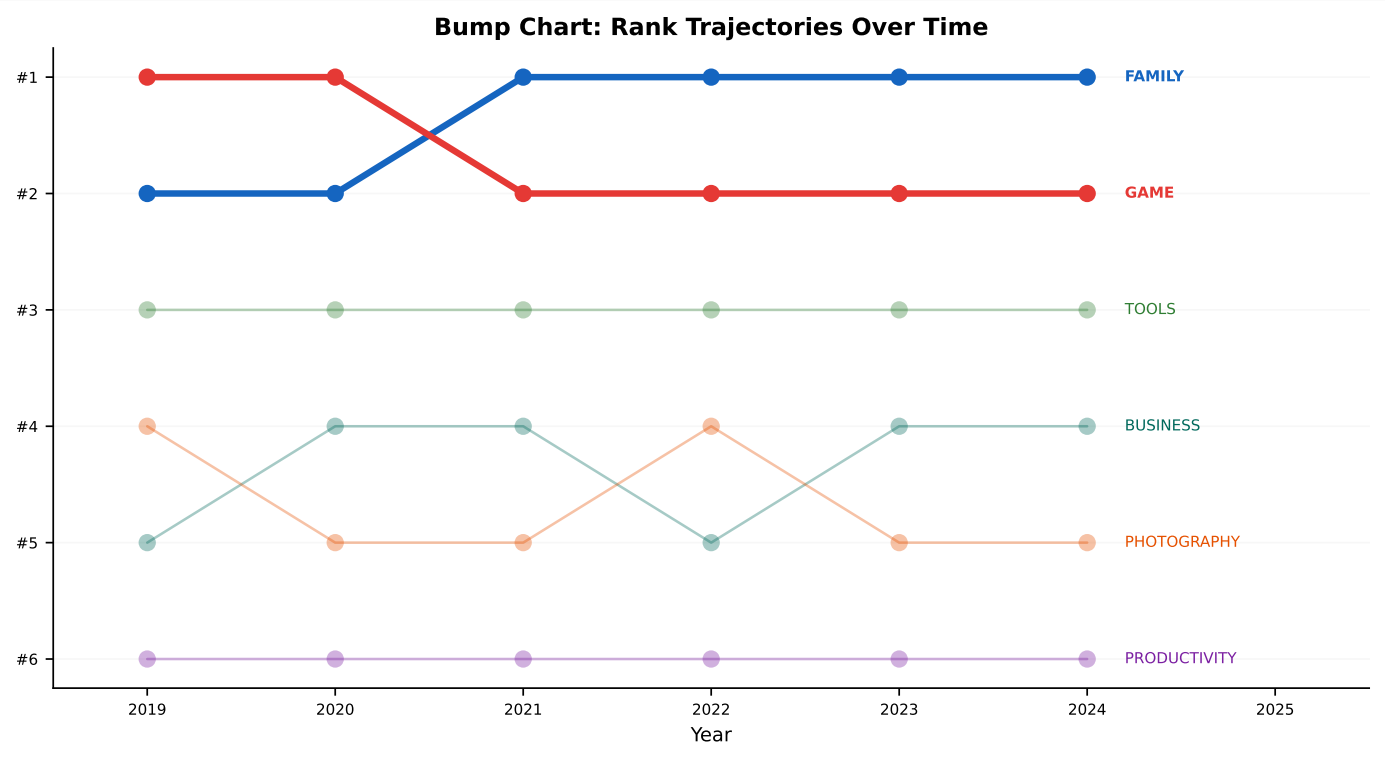
```r
df_long <- df |>
  pivot_longer(c(v2020, v2024),
    names_to = "year",
    values_to = "value") |>
  mutate(year = parse_number(year
      ))

ggplot(df_long, aes(
    x = year, y = value,
    group = category,
    color = highlight)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_text(data = . |>
    filter(year == 2020),
    aes(label = paste(category,
      scales::comma(value))),
    hjust = 1.1, size = 3) +
  geom_text(data = . |>
    filter(year == 2024),
    aes(label = paste(category,
      scales::comma(value))),
    hjust = -0.1, size = 3) +
  scale_color_manual(values =
```

Python

```python
fig, ax = plt.subplots(
    figsize=(7, 5))

for cat, v0, v1, c in zip(
    cats, v2020, v2024, colors):
    # Grey-accent
    lw = 2.5 if cat == key else 1
    alpha = 1.0 if cat == key
        else 0.3

    ax.plot([0, 1], [v0, v1],
        "-o", color=c, lw=lw,
        markersize=6, alpha=alpha
            )

    # Direct labels at both ends
    ax.text(-0.08, v0,
        f"{cat} ({v0:,})",
        ha="right", fontsize=7,
        color=c)
    ax.text(1.08, v1,
        f"{cat} ({v1:,})",
        ha="left", fontsize=7,
        color=c)
```

Bump Charts

Bump Chart: Rank Trajectories Over Time



Bump Chart in R vs Python

R (ggbump)

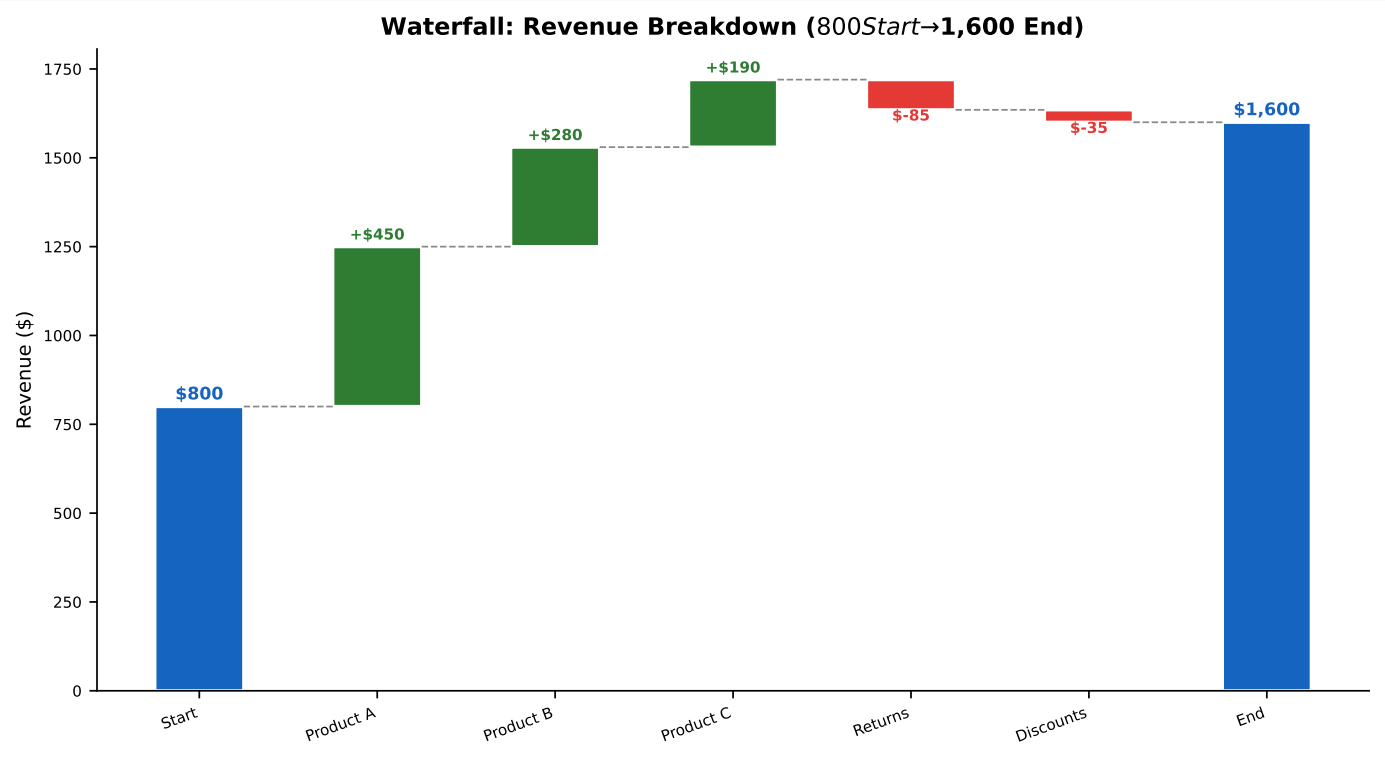
```r
# install.packages("ggbump")
library(ggbump)

ggplot(df_ranks, aes(
    x = year, y = rank,
    color = category)) +
  geom_bump(linewidth = 1.5,
    smooth = 8) +
  geom_point(size = 4) +
  geom_text(data = . |>
    filter(year == max(year)),
    aes(label = category),
    hjust = -0.2, size = 3,
    fontface = "bold") +
  scale_y_reverse() +
  scale_color_brewer(
    palette = "Set2") +
  theme_minimal() +
  theme(legend.position = "none",
    panel.grid.minor =
      element_blank()) +
  labs(title = "Category Rankings
      ",
       x = "Year", y = "Rank")
```

Python (manual)

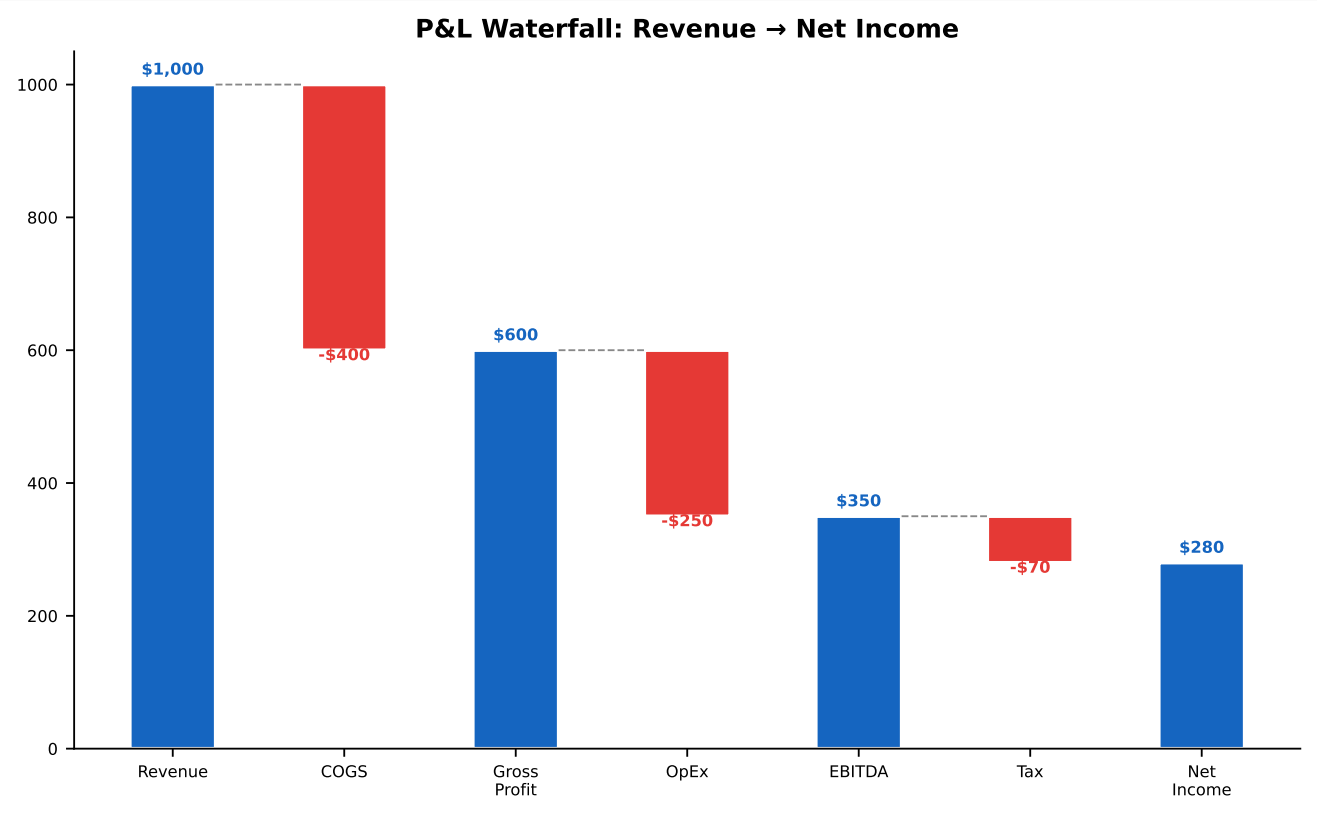
```python
fig, ax = plt.subplots(
    figsize=(9, 5))

for (cat, ranks), c in zip(
    rank_dict.items(), colors):
    # Grey-accent
    lw = 3 if cat in highlight
        else 1
    alpha = 1.0 if cat in
        highlight \
        else 0.3

    ax.plot(years, ranks, "-o",
        color=c, lw=lw, alpha=
            alpha,
        markersize=7)

    # Direct label at end
    ax.text(years[-1] + 0.2,
        ranks[-1], cat,
        fontsize=7, color=c,
        fontweight="bold",
        va="center")
```

Waterfall Charts

Waterfall: Sequential Cumulative Breakdown



P&L Waterfall: Revenue to Net Income



Data Insight

Waterfall charts decompose a total into sequential additions (green) and subtractions (red). The connector lines show the running total. Standard in financial reporting (P&L, budget variance). Start and end bars are blue (totals); intermediate bars are green (positive) or red (negative).

Waterfall in R vs Python

R (manual or waterfalls)

```r
# Manual waterfall with geom_col
df <- df |>
  mutate(
    end = cumsum(value),
    start = lag(end, default = 0)
        ,
    color = case_when(
      type == "total" ~ "total",
      value > 0 ~ "positive",
      TRUE ~ "negative"))

ggplot(df, aes(
    x = fct_inorder(item))) +
  geom_col(aes(y = value,
    fill = color),
    width = 0.5) +
  geom_segment(aes(
    xend = lead(item),
    y = end, yend = end),
    linetype = "dashed",
    color = "grey50") +
  scale_fill_manual(values =
    c(total="#1565C0",
      positive="#2E7D32",
```

Python

```python
for i in range(len(items)):
    if items[i] in ["Start","End"
        ]:
        ax.bar(i, running[i],
            width=0.5,
            color="#1565C0",
            edgecolor="white")
    else:
        bottom = min(running[i
            -1],
                     running[i])
        height = abs(values[i])
        color = ("#2E7D32"
            if values[i] > 0
            else "#E53935")
        ax.bar(i, height,
            bottom=bottom,
            width=0.5,
            color=color,
            edgecolor="white")

    # Connector line
    if i < len(items) - 1:
        ax.plot([i+0.25, i+0.75],
```

Chart Selection

Slope vs Bump vs Waterfall

Slope vs Bump vs Waterfall: When to Use Which

Chart	Time Points	Shows	Origin	Best For
Slopegraph	2 time points	Value change (absolute)	Tufte (1983) Edward Tufte	Best for before/after with 5–15 items
Bump chart	3+ time points	Rank trajectories (ordinal)	ggbump package sns + manual	Best for rank changes over 4–8 periods
Waterfall	Sequential additions	Running total (cumulative)	Finance, P&L budget breakdown	Start → contributions → end total

When to Use Each

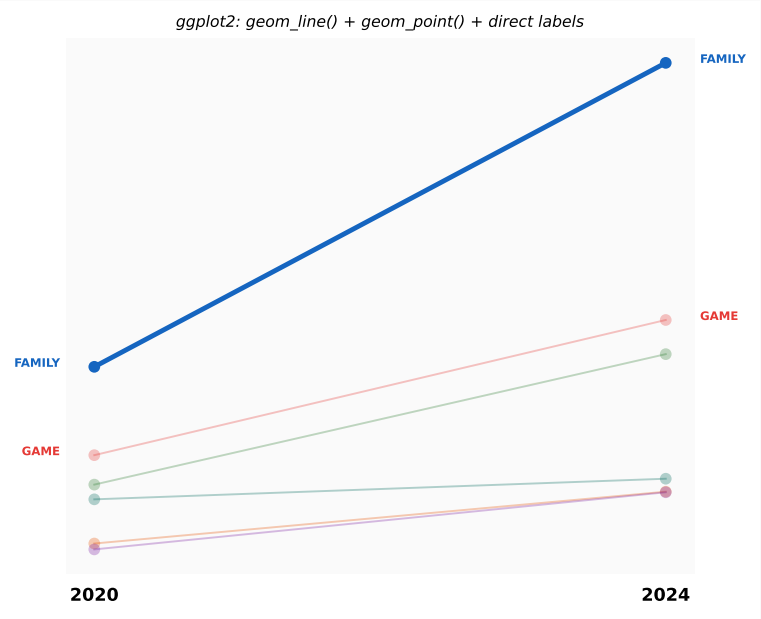
Scenario	Chart	Max Items
Before/after (2 points)	Slopegraph	5–15
Rank over time (3–8 periods)	Bump chart	4–10
Sequential breakdown to total	Waterfall	5–12 steps
Change comparison per category	Dumbbell (W03-M03)	5–15
Simple trend over time	Line chart (W06)	1–5 lines

Pro-Tip
All three charts benefit from the **grey-accent strategy**: render most items in light grey, highlight 1–2 in colour. This prevents spaghetti and directs the reader to the finding.

Complete Examples

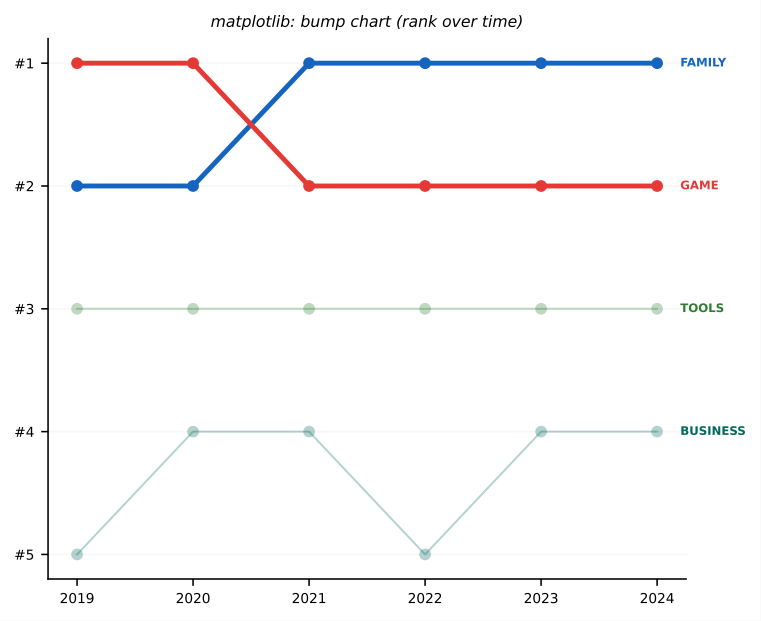
R: Slopegraph with Grey-Accent

```r
df_long |>
  mutate(highlight =
    category == "FAMILY") |>
  ggplot(aes(x = year, y = value,
    group = category,
    color = highlight,
    alpha = highlight)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values =
    c("TRUE" = "#E53935",
      "FALSE" = "#CCCCCC")) +
  scale_alpha_manual(values =
    c("TRUE" = 1, "FALSE" = 0.4)) +
  # Direct labels at both ends
  geom_text(data = . |>
    filter(year == 2020),
    aes(label = paste(category,
      comma(value))),
    hjust = 1.1, size = 3) +
  theme_void() +
  theme(legend.position = "none") +
  labs(title = "FAMILY Surged
    to #1 by 2024")
```



Python: Bump Chart

```python
fig, ax = plt.subplots(
    figsize=(9, 5))

for (cat, rk), c in zip(
    ranks.items(), colors):
    # Highlight top 2
    hl = cat in ["FAMILY", "GAME"]
    ax.plot(years, rk, "-o",
        color=c,
        lw=2.5 if hl else 1,
        alpha=1.0 if hl else 0.3,
        markersize=7 if hl else 4)
    ax.text(2024.2, rk[-1], cat,
        fontsize=7, color=c,
        fontweight="bold"
            if hl else "normal",
        va="center")

ax.invert_yaxis()
ax.set_yticks(range(1, 7))
ax.set_yticklabels(
    [f"#{i}" for i in range(1,7)])
ax.set_title(
    "FAMILY overtook GAME in 2021",
    fontweight="bold")
```



Synthesis

Key Takeaways

1. **Slopegraphs** show value change across exactly two time points. Slope = rate of change; crossing = rank reversal. 5–15 items max.
2. **Bump charts** track ordinal rank trajectories over 3–8 periods. Invert the y-axis so #1 is at the top. 4–10 items max.
3. **Waterfall** charts decompose a total into sequential additions (green) and subtractions (red) with connector lines. Standard for P&L and budget variance.
4. All three charts require the **grey-accent strategy** to avoid spaghetti when items exceed 5–6.
5. In R: manual `geom_line + geom_point + geom_text`, or `ggbump` for smooth bump curves. In Python: manual `ax.plot()` loops.

Essential Readings

- Tufte, E. R. (1983). *The Visual Display of Quantitative Information*, pp. 158–161 (Slopegraphs).
- Schwabish, J. (2021). *Better Data Visualizations*, Chapter 5 (Slope Charts, Bump Charts).
- Few, S. (2006). *Information Dashboard Design*, Chapter 6 (Waterfall charts in financial context).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 11 (Visualizing Nested Proportions).

Next Module: Uncertainty — Error Bars, Confidence & Gradient (W03-M07)